

A set of indicators for BPM life cycle improvement

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Abstract— Improving business processes and maintaining acceptable levels of performance are main factors in the success of any organization. Related to BPM (Business Process Management) [1] approaches, a variety of methodologies and techniques are available. These approaches include business process improvement methodologies (BPIM) [2] and process mining [3]. Actually, with BPM life cycle a continuous methodology of improvement business processes is defined, starting by the design phase and ending with the optimize phase. In this paper, we present a new vision BPM life cycle whose content differs from the classical one [4]. On each phase, we define a set of improvement indicators. These indicators recast business processes for best optimization results.

Keywords— BPM; BPM life cycle; Key Performance Indicators; Business Process; Performance Metrics; Process Mining

I. INTRODUCTION

As a result of the massive development induced by information and communication technologies, organizations are subject to fast changes in all aspects such as technical, organizational, and operational aspects. These changes need systems and procedures to increase the efficiency and performance of the business processes.

Business Process Management (BPM) makes an incremental improvement in business process, through several methodologies [2], techniques [5], and tools [6], and thus to ensure organization competitive position in the market.

According to [7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18], there is currently no integrated method to evaluate business process in term of defining improvement metrics for each BPM phase. Due to a challenging problem emerged, for companies wanted to improve their business processes during all phases of BPM, and for researchers evaluating new methodologies, techniques in business process improvement developments. Indeed, there is a need for an objectively way to checking improvement during the whole BPM life cycle phases, in terms of what can be optimized and how can be improved with each passing from one phase to another. This research fills this gap by defining a method based on a set of indicators for each phase.

This paper aims to define a set of improvement indicators for each phase of the BPM life cycle in order to improve the outcome business process.

This paper is divided into 4 mains sections planned as follows:

Section 2 presents a literature review of existing methods for process improvement and shows clearly the need for a method that covers the whole BPM lifecycle. In this section we also expose the proposed BPM life cycle. Section 3 presents the improvement indicators along the BPM life cycle. And section 4 concludes the paper and gives directions for future works.

II. BPM OVERALL

In this section, we introduce our new vision of BPM life cycle and the elements influencing it: the improvement metrics attributed to each phase, the outcome process criteria that must be obtained as a result of criteria applied to each outcome process and the specification format of the outcome process obtained from each phase of this BPM life cycle.

A. Literature review

The literature has defined several methods to improve business processes relate of the BPM life cycle (Table 1):

In [1], the authors focus on how redesigning business processes by presenting four dimensions to measure process performance quantitatively (time, cost, quality and flexibility), therefore, we observed that authors did not treat the three first phases (Design, Model, Execute) of BPM life cycle. further in [4], the authors define metrics only in the monitor phase by controlling data related to business process., so, this study doesn't support the 5 phases of BPM life cycle. Then as stated in [7], the authors present a scientific method to evaluate BPM products by defining a set of criteria for each BPM phase except the optimization phase. Also, as reported in [11] the author focuses on the objectives of performing BPM design. As well as cited in [14], the authors identify the new functional requirements for a Semantic Business Process Management (SBPMS) for each phase of the BPM lifecycle and explain the benefits of adopting semantic technologies in SBPM. The authors specify requirements, rather than solutions and metrics. Thus [16], the authors proposed a method to evaluate and monitor the business process against performance requirements and show the effects of ongoing processes on business goals, in a real-time manner. Additionally, in [17] the authors treat only the implementation phase of BPM life cycle. Furthermore in [18], the authors focused on using data warehouse in the BPM life cycle for supporting the tree following phases (Execute, Monitor, Optimize). While only [19], considers the whole BPM life-cycle by implementing techniques for process mining and intelligent redesign to

support the (re)design and diagnosis phases and thus close the BPM life-cycle.

According to the table 1, there is no integrated method to evaluate business process in term of defining improvement metrics for each phase of the BPM life cycle.

TABLE I. BPM METHODS SUMMARY

Methods	Supported phases
Dumas et al. [1]	Redesign
Classical BPM [4]	Monitor
Koster [7]	Discovery, Modelling, Design, Deployment, Execution, Monitoring, Analysis
Scott Simmons 2013 [11]	Design
Semantic Business Process Management (Required functionalities) [14]	Modelling, implementation, Execution, Analysis
Business Process Monitor & Alignment [16]	Monitor, optimize
Grey-based DEMATEL Model (Factor implementation) [17]	Implementation
Evaluation Framework for Business Process Evaluation Approaches (Implementing Data warehouse) [18]	Execute, Monitor, Optimize
[19]	(Re)Design, Diagnosis

B. BPM vision

To improve business processes, we based on the BPM life cycle, that can be involved change in existing systems, teams, or processes.

As shown in figure 1, identify the relevant business of a company is the first step in this new vision, which help to discover its “as-is” process: (defines the current state of the business process in an organization.), then by applying the design metrics on it, we will obtain its “to-be” process (defines the future state of a business process in an organization) as an outcome of the design phase. In turn, it can be analyzed to acquire the definition of execution metrics, based on results of the model phase and the SLAs [20] (Service Level Agreement) of this company.

After execution, we have as an outcome an executable process that must be monitored by a set of KPIs (Key Performance Indicators) in the monitor phase. As a result, we get the monitored process that must be optimized by applying the process mining techniques as a last step of this BPM life cycle.

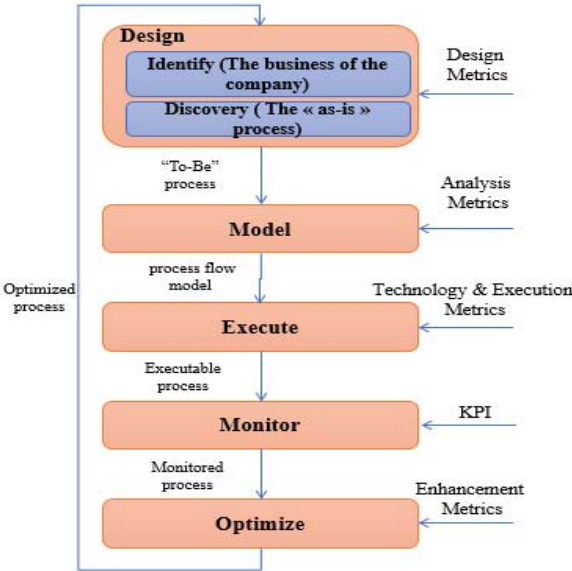


Fig.1. BPM life cycle metrics

C. Advantages of our BPM vision

The advantage of this vision over the classical one [4] (define metrics as KPIs only in the monitor phase) revolves around the improvement of business processes by the contribution of improvement indicators. These indicators are presented as different metrics (see Figure 1) applied in each phase of the BPM life cycle, in order to obtain optimized business processes (see Table 2).

TABLE II. OUR BPM VISION’S BENEFITS.

Phases	Our BPM vision	Classical BPM [4]
Design	Design Metrics	No metrics
Model	Analysis Metrics	No metrics
Execute	Execution & Technology Metrics	No metrics
Monitor	KPIs	KPIs
Optimize	Enhancement Metrics	No metrics

III. BPM PHASES & IMPROVEMENT STEPS

The improvement metrics are attributed to each phase of BPM life cycle, in order to obtain a specific format of outcome processes during the whole phases of the BPM life cycle.

1) Phase 1 :Design (Design metrics)

In this phase, we are faced to two parts: Identify the Business of the company and Discovery the “as-is” process (see Figure 1).

We define the set of indicators related to the design phase according to a specific logic. We start by defining the Business of the company (Relevance), that leads to check the syntax of the process(Correctness). Once this Correctness is verified, it become easy to understand the language representing the process (Clarity). Based on the clarity of the

process, we can complete the process throughout its implementation (Completeness), then, we focus on representing it relatively to reality (Consistency), that emerged to obtain the “to-be” process. Throughout this logic, we propose the order of priority as presented in table 3 as follow (Relevance → Correctness → Clarity → Completeness → Consistency) [21].

TABLE III. DESIGN METRICS

Order by Priority	Design Metrics
1	Relevance
2	Correctness
3	Clarity
4	Completeness
5	Consistency

So, these improvement indicators described in this phase allow to have a good decision diagram: Have correct execution semantics, lay out a clear structure that employs sub decisions if necessary, provide complete mapping of all possible input/output combinations, employ data objects that are consistent with the real-life data.

2) Phase 2: Model (Analysis metrics)

In this phase we define a set of indicators named analysis metrics (see Table 4). These indicators will be applied on the “to-be” process, that we retained from the previous phase of our BPM life cycle and conduce to obtain the “process flow model” (The outcome of this phase is a pseudo-executable *process flow model* that can be exported as an *executable language format* supported by the BPM engine (such as ORACLE BPM, Orchestra ...). Based on the mapping from a modelling language (as BPMN) to a process execution language (as BPEL...) compatible with the BPM engine).

These analysis metrics based on the logic combined with, the order of need for each part of the formulas to another, for instance we must calculate previously the cycle time to obtain the throughput rate. The values are comparable to the company’s business objectives and verifiable to the outcome process criteria: (if the BP: **1. Valid, 2. Efficient, 3. Effective, 4. Economic, 5. Usable** [22]).

In [22], the authors focus on what would make such a process a good process model, and during our BPM life cycle we want at the end of each phase to check if the process achieves these good process model conditions. These conditions are distributed according to the requirements of each phase, that we will present in the following BPM phases.

TABLE IV. ANALYSIS METRICS (IN SIMULATION)

Order by priority	Analysis Metrics	Formula
1	Operation time	Setup time + Run time
2	Cycle time	Average (time units)
3	Throughput rate	1 / cycle time
4	Efficiency	Value Added Time / cycle time
5	Productivity	Output / Input
6	Utilization of an Operation	Demand / Capacity

The analysis metrics we retained from literatures [23] are defined as follow:

Operation time: **Setup time** (the length of time required to changeover from one process/activity to another) + **Run Time** (is when a process is running (or being executable). That is, when we start a process running in a computer, it is runtime for that process).

Throughput time: Average time for a unit to move through the system or the amount of time a flow unit spends in a business process from beginning to end. If there is more than one path through the process, the flow time is equivalent to the length of the longest path.

Cycle Time: is the average successive time between completions of successive units.

Throughput rate: The number of flow units (e.g. customers, money, produced goods/services) going through the business process per unit time, e.g. served customers per hour or produced parts per minute. The flow rate usually is an average rate.

Process Cycle Efficiency: This is a measure of how much waste is present in your process.

Productivity: the ratio between the amount produced and amount of resources used in production. Units of Input means resources utilised, and Units of output means anything generated from production: (Land: Hectares, Metric: ton, Machinery: machine hours and People: Man Hours...).

Utilization: Is the ratio of the time that a resource is activated relative to the time that it is available for use.

Capacity: maximum output of a process or resource measured in units/time: a rate.

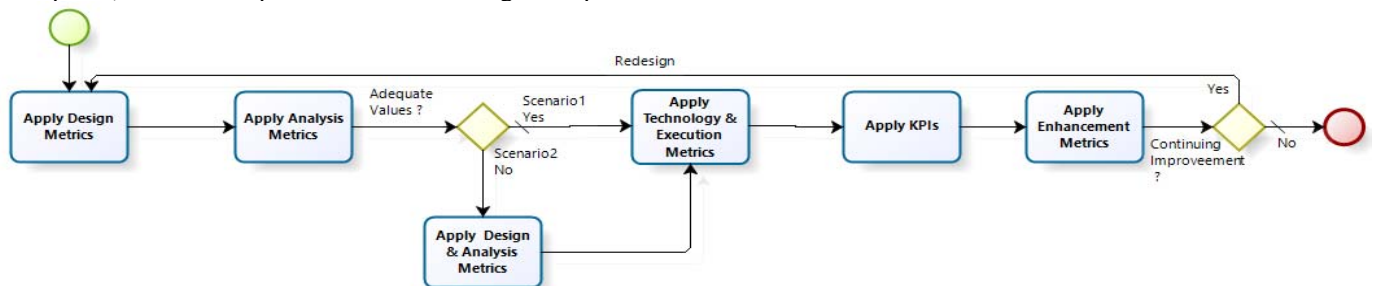


Figure 2. The model phase’s scenarios

Synthesis: These analysis metrics allow to evaluate the business processes of a company before their final use/employment. After applying the analysis metrics on the “to-be” process in simulation step, Simulation is a method for visualizing and validating the behaviour of our business process diagram with visual indications of all currently simulating activities along with the possible activities that can be executed next, we will easily be able to understand and identify potential issues with the process we have modelled, we will be facing several scenarios (see Figure 2). In scenario 1, after applying the analysis metrics on the “to-be” process, we suppose a change in notation on the process model (E.g. the BPMN diagram) and in the resource charge.

3) Phase 3: Execute (Execution and Technology metrics)

In this phase, and after the simulation step, we arrive at the execution of the process flow model in real time, where we will define two classes of metrics (see Table 5): Technology metrics and Execution metrics. We will apply these metrics to the current system to get an executable process: **1. Measured and 2. Managed** [22]. In the table 5, these metrics are based on the order:

For both metrics: To have the executable process, the most suitable technology must be prepared, in which this process should be implemented, as well as the metrics allow it to be executed in good conditions according to the company.

For Technology metrics:

The order is given from global to specific (System→ Network→ Support application).

For Execution metrics: perform the correspondence between Time spent (references & frequencies), the most optimal cost and quality according to our business process.

Technology Metrics [24]:

Uptime: The percentage a system is “running”, and downtime is the inverse of that. So, the key indicator to follow it is “availability”, which is a measurement of whether a system’s, application’s or network’s target users can access and normally use a system. A system can be “up” but unavailable, therefore, when defining SLAs or “working agreements” with businesses, it is best to focus on availability guarantees and not just simple uptime.

Execution Metrics [24]:

Time Reference (for current or former instances): Through time (min, max, average...). Cycle time (min, max, average...).

Processing time (duration of process step) of all subject carriers (min, max, average...).

Frequencies:

For running instances: Number of open process instance per process type. Number of open process instances per process type in which a subject carrier is involved. Start time of instances per type of process (distribution). Number of instances, which are in a process step. Number of instances that have exceeded a certain period a particular transition).

For completed instances: Number of process instances per process type and time unit. Number of process instances for which the throughput time is above average. Number of subject change of a subject carrier per time unit.

Costs: Total cost of the process. Cost of a subject carrier in the process. Propositions of subject carriers to the total cost of the process.

Quality: Numbers of mistakes in result. Proposition of reworking correction errors.

TABLE V. TECHNOLOGY & EXECUTION METRICS (IN REAL TIME)

Metrics	Order by priority	Indicators	Finality
Technology Metrics [24]	1	System Uptime	According to the company's measures
	2	Network Uptime	According to the company's measures
	3	Support application Uptime	According to the company's measures
Execution Metrics [25]	1	Time reference	Detection of variation and exploration of causes
	2	Frequencies	Identification of bottlenecks in total and exploration of causes
	3	Cost	Identification of the quantity structure and exploration of causes
	4	Quality	Identification of the faulty process results and classification of errors

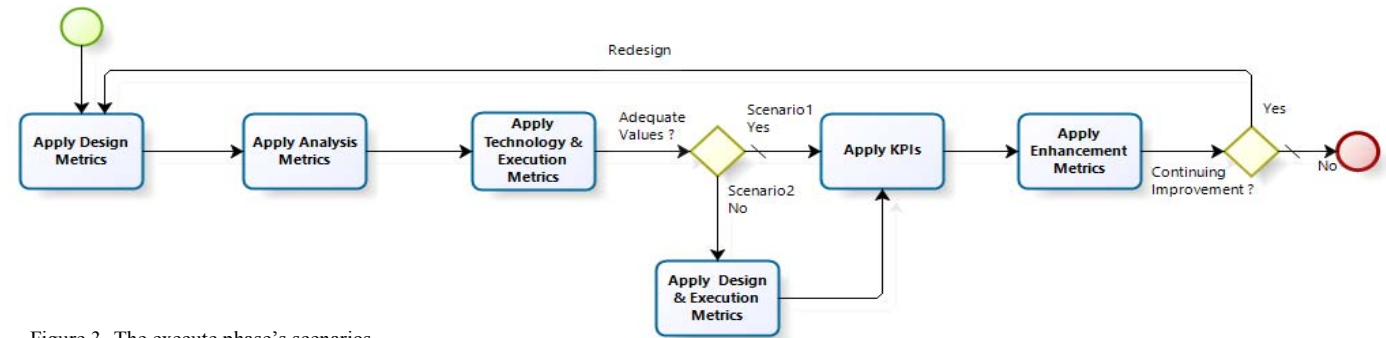


Figure 3. The execute phase’s scenarios

Synthesis:

In the fact, the process flow model is executed, then the values obtained are compared with the SLAs and the company business objectives. In the case of a discrepancy between these values and based on these metrics, some changes can be suggested to, the system's properties or, the process model (see Figure 3).

4) Phase 4: Monitor (KPIs)

In the fact, after the process has been running for a while in the execution phase, and all traces have been collected, we periodically monitored these informations, for obtaining a monitored process model: **1. Used and 2. Reused** [22].

According to [26] concluded that the outcomes most studied in the manufacturing field (quality, cost, flexibility and delivery). The research work [27] find that the performance outcomes of business processes defined by (reduced cost, reduced time, improved quality, improved work life quality and enhanced learning and empowerment). Then, the contribution in [28] cited eight dimensions of performance features (reliability, conformance, durability, serviceability and perceived quality). The service quality domain proposed five attributes of quality (service as reliability, responsiveness, assurance, empathy and tangibles [29]). So, Performance measure should include how well process is working, how is the process influencing the company's environment as handling changes. Therefore, the performance measures can be determined as process efficiency process effectiveness and process flexibility.

The authors in [30] detect the link between some of the previous metrics used to measure the performance and these three meta-measures.

Based on [26, 27, 28, 29, 30] and [31], we grouped our KPIs into (see Table 6): process efficiency, process effectiveness and process adaptability. We define our KPIs based on the logic combined with: (Calculate time, cost, utility of the BP which leads to check if it is flexible relatively to user satisfaction and with suitable quality).

KPI/ Oder by priority	Dimensions / Metrics			Finality
Efficiency (1)	Cost: -Material Equipment -Opportunity cost -Cost of poor quality -Cost added only (rework, waste, duplication) -Life Cycle Cost	Time: -Cycle Time -Task Time Up Time -Down Time -Delay Time	Utility: -Personal Turnover -System uptime -Network uptime -Support application uptime	-The performance measures associated with the process with measures related to the process itself
Adaptability (2)	Flexibility [32]: - Durability -Responsiveness - Timeliness			-The ability of the process to adapt to the change
Effectiveness (3)	Satisfaction: -Requirement of stakeholders are met -Rating of service/product by importance and satisfaction -Employee engagement rating -New customers attracted -Retention rate of current costumers/employees - Customers/ employee complaints	Quality: -Accuracy -Uniformity -Reliability -Deliverability		-The performance measures associated with the process with respect to the change induced to the environment.

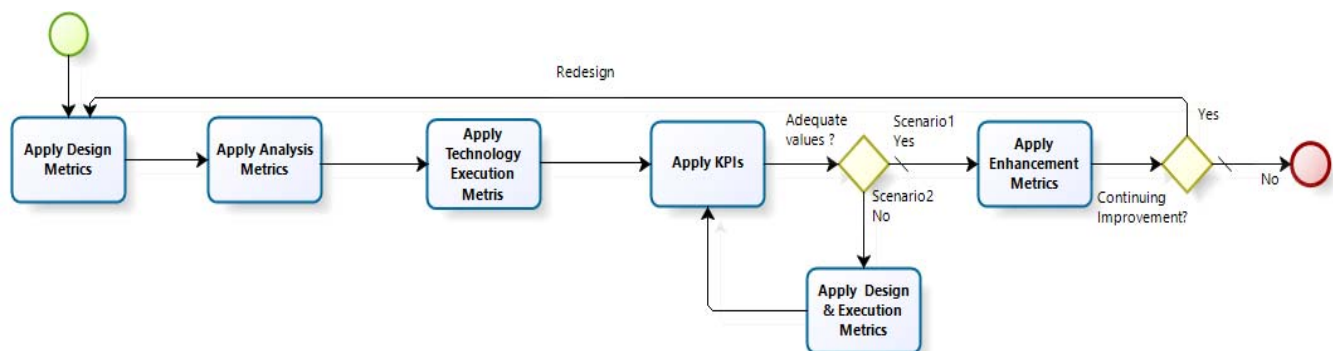


Figure 4. The monitor phase's scenarios

TABLE VI. KPIs

Synthesis: After applying the KPIs, we suppose that we detect some bottlenecks (delay, dissatisfaction...) in the results obtained. To use these results in improvement cycle of our supposed business process, we suggest two scenarios (see Figure 4). We may modify the process flow model (see Figure 4) for adapting these metrics to bottlenecks occurred on the monitor phase. Then, passing to the next phase. Or Passing directly to the process mining phase (see Figure 4) in order to redesign the process flow model, and to adjust metrics, as a solution to obtained errors.

5) Phase 5: Optimize (Enhancemnt Metrics)

In this phase, we focus on how to use the process mining (PM) in our BPM life cycle, and with which its techniques will be involved during this life cycle.

After the process has been running for a while in the execute phase and “event logs”, collection of cases, where each element refers to a case, an activity and a point in time, have been collected. We use “process mining” which consists of a set of analysis techniques that can be used to understand the actual behaviour and performance of business processes by studying events recorded in IT systems techniques [33] (also known as event logs), on these collected event logs by applying: Quality Criteria, Control flow perspective, organizational perspective, Case perspective and time perspective. To up end with an optimized process that must be: **1. Adaptable 2. Efficient 3. Effective** [22].

Table 7 illustrates five types of metrics used in the process mining architecture [34], these metrics are classified with the order (Discovery → Conformance → Enhancement). Quality Criteria are: Fitness, Precision, Generalization and Simplicity. The process mining perspectives used are grouped as follow: the control-flow perspective focuses on the control, i.e., the ordering of activities. The organizational perspective focuses on information about resources (e.g., people, systems, roles and departments). The case perspective focuses on properties of cases, e.g., cases can also be characterized by the values of the corresponding data elements.

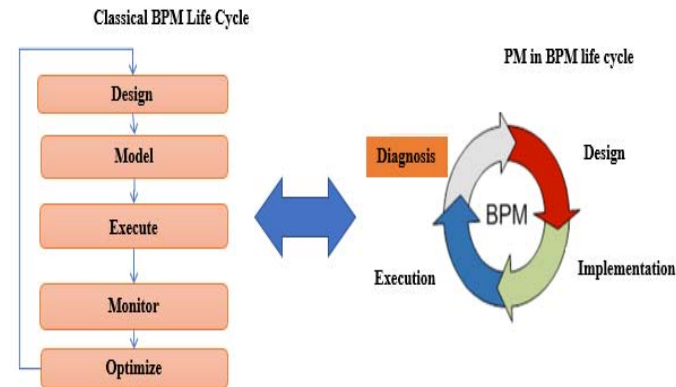


Figure 5. BPM life cycle

Synthesis: We consider process mining as the last phase in our BPM life cycle. for this reason, it is necessary to introduce the positioning of these techniques in the BPM life cycle. Therefore, we will focus on the link between BPM and PM. The BPM life cycle illustrated in figure 5 is describes the correspondence between the classical BPM life cycle and the appropriate one for process mining [19]. Although, figure 5 shows how iterates through multiples phases in designing (Design), implementing (Model), executing (Execute), analyzing (Monitor & Optimize), then re-designing the processes. While classical BPM approaches start with modeling phase, the process mining starts by discovering the processes that are presented by the event logs.

So, Process mining clearly positioned in the Optimize phase of the BPM life-cycle.

By passing with process mining techniques, we obtained the latest optimized process “enhanced process”. Therefore, Process mining establishes links between the actual processes.

TABLE VII. ENHANCEMENT METRICS

Order by priority	Metrics	Finality
1	Quality Criteria [35]	1. Fitness 2. Precision 3. Generalization 4. Simplicity
2	Control flow Perspective [36]	Focuses on the ordering of activities, in order to find a good (can be expressed in Petri net or some other notation as EPCs, BPMN, UML etc.).
3	Organizational Perspective [37]	Focuses on the information about resources hidden in the log, in order to structure the organization by classifying people.
4	Case Perspective [38]	Focuses on properties of cases
5	Time perspective [39]	It's concerned with the timing and frequency of events. It makes possible to discover bottlenecks, measure service levels

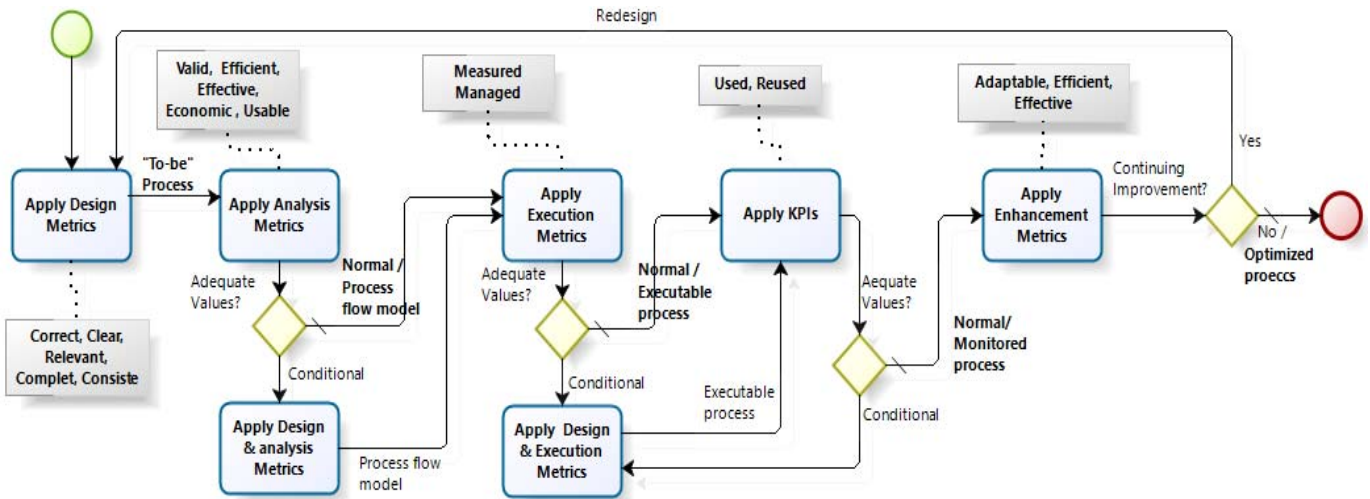


Figure 6. BPM life cycle cases

IV. CONCLUSION

In this paper, we focused on supporting the whole BPM life cycle consisting of the following phases: Design, Model, Execute, Monitor, Optimize.

First all, we have proposed a new vision of BPM life cycle with the following elements: metrics, outcome process, outcome process criteria. The metrics present the different measures attributed to each phase in order to improve the business processes during its passage from one phase to another: Design metrics, analysis metrics, execution metrics, KPI and enhancement metrics. The phases encompass all the different phases of our BPM life cycle: Design, Model, Execute, Monitor, optimize. Outcome process shows the outcome of each phase which is represented in the form of a specific process: "as-is" process, "to-be" process, process flow model, executable process, monitored process, optimized process. Outcome process criteria [22] illustrates the set of criteria to be checked on each outcome process at the end of each phase: Valid, relevant, correct, clear, complete, effective, efficient, usable...

Second, we observed two cases of BPM life cycle (see Figure 6): Normal & conditional. Also, figure 6 shows the original BPM life cycle combined with the logic of our new vision by adding: metrics, outcomes processes and outcome processes criteria. The normal is specified by the following path: Apply the design metrics → Apply the analysis metrics → Apply execution & technology metrics → Apply KPIs → Apply enhancement metrics. While, the conditional one summaries how metrics are affected the business process in specific cases for obtaining the suitable business process by following the **conditional path** in figure 6.

Lastly, we have presented best metrics to improve business process, and best outcome process criteria to have a good process.

As perspective, more metrics [40, 41] need to be defined to build a completed tool for BPM life cycle improvement, and we are working on a complete case study to validate our proposition.

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